

Prism Surface Area Wrap

Unit 10: Time Capsule Engineers • Lesson 10-4 • EduWonderLab

UNIT 10 • LESSON 10-4 • TEACHER NOTES

Standard 6.G.4

FORMAT Face-pairing task cards

TIME 25–35 min

GROUPING Pairs

TOPIC Surface Area of Prisms

STANDARD 6.G.4

PREP LEVEL Light — print prism cards and wrapping organizer

Learning goal *I can find the surface area of a rectangular or triangular prism.*

Teacher setup

- Print one set per pair. Students “wrap” each prism by finding the area of every face.
- They organize faces into pairs and add for the total.
- Include rectangular and triangular prism rounds.

Materials

Prism cards, Wrapping-paper organizer, Face labels, Pencil.

Differentiation & language support

- Organizer groups matching faces in pairs (SPED).
- Sentence stem: “I found the area of all ____ faces and added them.”
- ESOL: label faces top, bottom, front, back, sides.

Key vocabulary (English / Español):

Term	Español	What it means
face	cara	a flat surface of a solid
lateral area	área lateral	the area of the side faces

Quick teacher check: *Check the lateral area on Card 5 (perimeter $12 \times$ length $8 = 96$).*

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Name: _____

Date: _____

Class: _____

Your goal: *I can find the surface area of a rectangular or triangular prism.*

What you need

Prism cards, Wrapping organizer.

How to play

1. Pick a prism card and identify each face.
2. Find the area of each face.
3. Add all the face areas.
4. Record the surface area in square units.

Prism Cards

Find the area of every face, then add. Opposite faces of a box come in matching pairs.

1. Rectangular prism $4 \times 3 \times 5$	2. Rectangular prism $6 \times 2 \times 3$
3. Triangular prism: triangle base 6, height 4, sides 5, 5, 6; length 10	4. Cube $2 \times 2 \times 2$
5. Triangular prism: right triangle legs 3, 4, 5; length 8	6. Rectangular prism $8 \times 3 \times 2$

Wrapping organizer (face areas + total):

Explain your thinking

Explain why a rectangular prism's faces come in three matching pairs.

Sentence frame: *I found the area of all ___ faces and added them to get ___.*

★ **Challenge:** *For Card 3, identify the two triangular faces and the three rectangular faces separately.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Rectangular prism $4 \times 3 \times 5$ Answer: 94 sq units $[2(12 + 20 + 15)]$	2. Rectangular prism $6 \times 2 \times 3$ Answer: 72 sq units $[2(12 + 18 + 6)]$
3. Triangular prism: triangle base 6, height 4, sides 5, 5, 6; length 10 Answer: 184 sq units $(24 + 160)$	4. Cube $2 \times 2 \times 2$ Answer: 24 sq units (6×4)
5. Triangular prism: right triangle legs 3, 4, 5; length 8 Answer: 108 sq units $(12 + 96)$	6. Rectangular prism $8 \times 3 \times 2$ Answer: 92 sq units $[2(24 + 16 + 6)]$

Teaching notes & look-fors

- Box surface area = $2(lw + lh + wh)$. Triangular prism = 2 triangles + 3 rectangles.
- Card 5: triangles $2 \times 6 = 12$; lateral $12 \times 8 = 96$; total 108.